03 | Top-Tier Product Companies

Advanced software engineering interviews - algorithms and problem solving

Target interview difficulty: Mostly Medium, with selected Hard problems

COMPANIES

Google, Meta, Amazon, Microsoft, Apple, Netflix, Adobe, Uber, Salesforce, LinkedIn, Atlassian, Sprinklr, X (Twitter), Texas Instruments

DSA TOPICS

All core DSA: Arrays, strings, hashing, linked lists, stacks, queues, heaps, binary search, trees and BST.

Interview patterns: Two pointers, sliding window, prefix sums, intervals, greedy, recursion and backtracking.

Graphs & advanced structures: BFS, DFS, topological sort, shortest paths (Dijkstra), union-find, trie; segment tree when relevant.

Advanced algorithms: Dynamic programming (1D/2D), bit manipulation, selected number theory, advanced graph problems.

INTERVIEW PREPARATION

• Practice Medium problems deeply, then targeted Hard problems; prioritize clarity over memorizing solutions.

• Discuss alternative approaches, correctness, complexity, test cases and trade-offs.

• Typical practice: word break, course schedule, LRU cache, longest increasing subsequence, graph shortest path, trie search and grid DP.

Note: A concise preparation guide based on your uploaded PDFs. These are suggested study priorities, not guaranteed company-specific questions.